KneeGait-Edge

Democratizing Gait Intelligence

A ₹516 Edge-AI Wearable for Accessible Gait Analysis

₹516 ACCESSIBLE BY DESIGN



THE ACCESS GAP

Gait analysis shouldn’t be locked behind expensive infrastructure.

Traditional Approach

- Specialized equipment
- Expensive infrastructure
- Laboratory-dependent
- Limited access



The Need

Portable • Affordable • Real-time • Accessible

Can gait intelligence become portable, accessible and affordable?

THE PROTOTYPE

Meet KneeGait-Edge

A wearable Edge-AI prototype measuring dynamic knee flexion using two IMUs mounted on thigh and shank.

01 SENSE
Two IMUs capture movement

02 COMPUTE
ESP32 processes in real time

03 ANALYZE
Gait features are extracted

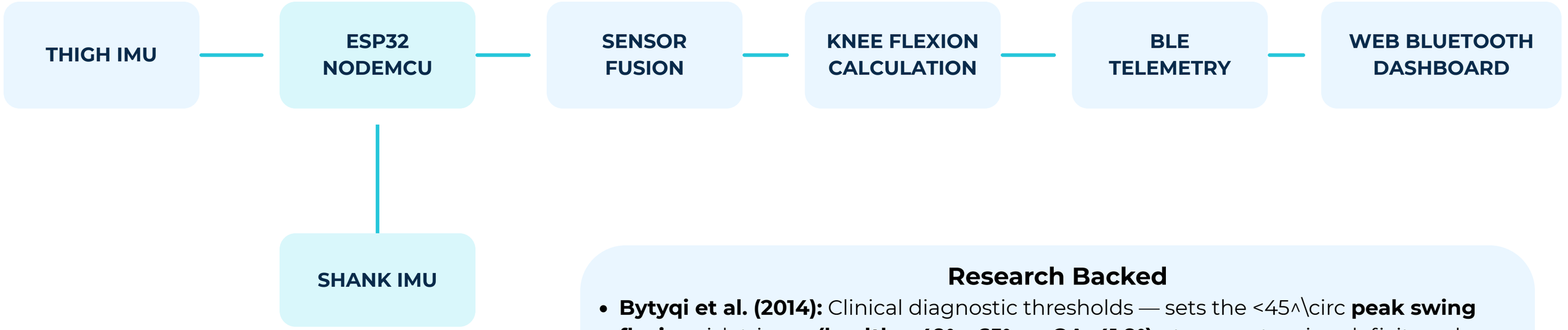
04 VISUALIZE
BLE streams to the browser



From body movement to real-time gait intelligence.

SYSTEM FLOW

From Motion to Gait Intelligence



Research Backed

- **Bytyqi et al. (2014):** Clinical diagnostic thresholds — sets the <45^\circ **peak swing flexion** risk trigger **(healthy: 49° – 65° vs. OA: 41.9°)**, stance extension deficit, and sagittal Range of Motion (ROM).
- **Rhudy et al. (2024):** Core sensor fusion algorithm — provides the magnetometer-free **complementary filter (α = 0.96 at 50 Hz)** and validates dynamic joint-angle tracking within **RMSE ≤ 3.2°**.
- **Song & Baxter (2025)**: Hardware placement & validation baseline — dictates mounting the IMU to the **anteromedial tibia** to eliminate muscle wobble, validating wearable IMUs against optical motion capture with **Rxy ≥ 0.90** correlation.

HARDWARE

Simple Hardware. Intelligent Processing.

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PROTOTYPE COST

ESP32 NodeMCU
Processing + BLE

2 × MPU-6050
6-axis motion sensing

USB Power Bank
Portable power

Elastic Bandage
Wearable mounting

I²C BUS

```
SDA → GPIO 21      SCL → GPIO 22      Thigh IMU → 0x68      Shank IMU → 0x69
```

Cost democratization : capable sensing with components accesible by everyone.

SIGNAL FUSION

Turning Raw Motion Into Knee Flexion

01
ACCELEROMETER TILT
Orientation from acceleration

02
GYROSCOPE INTEGRATION
Dynamic rotation

03
COMPLEMENTARY FILTER
Fused signal, reduced noise

$$\theta(t) = 0.96 \times \theta gyro(t) + 0.04 \times \theta acc(t)$$

Knee Flexion = |θthigh − θshank|

50 Hz sampling • α = 0.96

EDGE AI

Intelligence at the Edge

RAW 50 Hz
MOTION DATA

STRIDE
SEGMENTATION

GAIT FEATURE
EXTRACTION

RANDOM FOREST
CLASSIFICATION

ON-DEVICE
INFERENCE

Peak Swing
Flexion

Flexion Range
of Motion

Stance / Swing
Ratio

Mean Angular
Velocity

Random Forest → portable C → ESP32 inference
No cloud dependency for core inference

LIVE EXPERIENCE

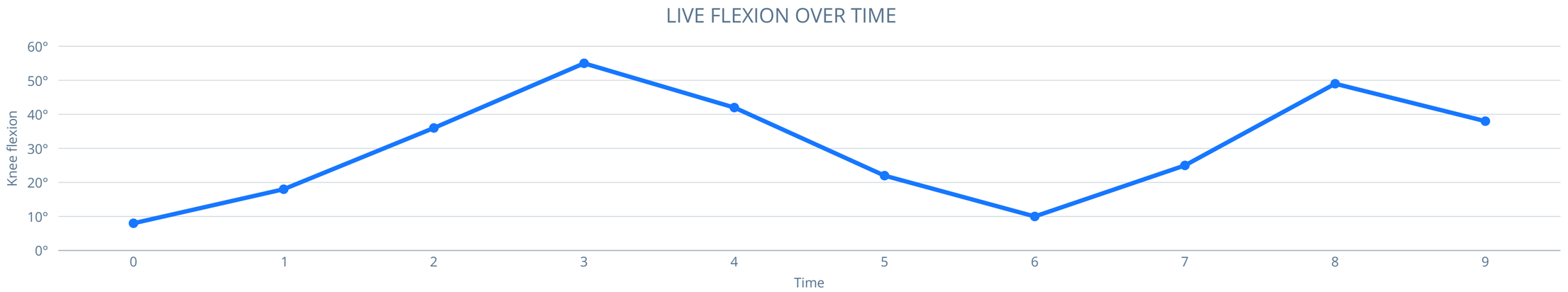
See the Gait. In Real Time.

CONNECTED / WEB BLUETOOTH

42.8°
DYNAMIC KNEE FLEXION

58.4°
PEAK SWING FLEXION

Clinical Evaluation Status
Screening view — research prototype



Illustrative UI examples; not patient results.

BIOMECHANICAL CONTEXT

From Movement to Meaning

≈ 55°–65°

Healthy peak swing flexion reference

<45°

Screening indicator for stiff-gait risk — not a diagnosis



STANCE TOE-OFF MID-SWING HEEL STRIKE

KneeGait-Edge is a research/hackathon prototype and is not intended to provide a medical diagnosis.

ACCESSIBILITY

Democratizing Gait Intelligence

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PROTOTYPE OUT-OF-POCKET COST

EXPENSIVE INFRASTRUCTURE

PORTABLE SENSING

EDGE COMPUTATION

ACCESSIBLE GAIT INTELLIGENCE

Low-cost components • Wearable form • ESP32 edge processing • BLE • Browser visualization • No dashboard backend

Advanced gait insights shouldn’t require advanced infrastructure.

ROADMAP

From Hackathon Prototype to Scalable Gait Intelligence

NOW

₹516 prototype
Dual-IMU sensing
ESP32 Edge AI
BLE dashboard

NEXT

Larger gait datasets
Robust model validation
Improved enclosure
Better signal processing

FUTURE

Clinical validation
Mobile integration
Long-term monitoring
Broader accessibility

POTENTIAL APPLICATIONS

Research • Physiotherapy support • Gait screening • Remote monitoring • Resource-constrained environments

CREDITS

References & Acknowledgements

RESEARCH REFERENCES

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Democratizing Gait Intelligence.

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Note : product images used are AI generated and for display purpose only, actual prototype is yet to be build